DATA QUALITY & CLEANING AUDIT REPORT

Dataset: sample_customer_churn.csv | **Generated:** 2026-09-24 19:57:44 | **Engine:** AutoCleanPro v2.0

INITIAL QUALITY SCORE	POST-CLEANING SCORE	NET IMPROVEMENT
91.4%	**100.0%**	**+8.6% Improvement**
Grade: A (Excellent)	Grade: A (Excellent)	Status: Fully Optimized

1. Data Health Comparison (Before vs. After)

Metric	Raw Dataset	Cleaned Dataset	Impact / Status
Total Rows	17	15	-2 rows change
Total Columns	10	10	Retained
Missing Cells	3 (1.76%)	0 (0.0%)	**Resolved**
Duplicate Rows	2 (11.76%)	0 (0.0%)	**Removed**
Outliers Identified	8	0	**Handled**
Completeness Sub-Score	98.2%	100.0%	+1.8%
Uniqueness Sub-Score	88.2%	100.0%	+11.8%

2. Automated Cleaning Pipeline Log

#	Cleaning Operation Performed
1	Removed 2 duplicate row(s).
2	Trimmed leading/trailing whitespace across 4 string column(s).
3	Converted 1 negative value(s) in column 'Total_Spend' to positive values.
4	Imputed 2 missing value(s) in numeric column 'Annual_Income' with median (65000.0).
5	Imputed 1 missing value(s) in numeric column 'Credit_Score' with median (690.0).
6	Clipped 1 outlier(s) in column 'Age' to IQR bounds [5.25, 71.25].
7	Clipped 1 outlier(s) in column 'Annual_Income' to IQR bounds [3500.0, 127500.0].
8	Clipped 1 outlier(s) in column 'Total_Spend' to IQR bounds [-2261.67, 5317.72].

3. Detailed Column Issues & Type Analysis

Column Name	Detected Data Type	Missing Count (%)	Outliers (IQR)	Suggested Fix
CustomerID	str	0 (0.0%)	0	Passed All Checks
Customer_Name	str	0 (0.0%)	0	Passed All Checks
Age	int64	0 (0.0%)	1	Cap/clip outliers
Gender	str	0 (0.0%)	0	Passed All Checks
Annual_Income	float64	2 (11.76%)	1	Impute missing, Cap/clip outliers
Credit_Score	float64	1 (5.88%)	0	Impute missing
Join_Date	str	0 (0.0%)	0	Passed All Checks
Total_Spend	float64	0 (0.0%)	2	Cap/clip outliers
Is_Active	int64	0 (0.0%)	0	Passed All Checks
Churn	int64	0 (0.0%)	4	Cap/clip outliers